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*Description, Objectives, Ethics & Resources*

# Predictive vs. Reactive Paradigms in Multi-Asset Trading

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# 1 Introduction

Algorithmic trading has become increasingly prevalent in modern financial markets, with machine learning models playing a central role in automating decision-making. For my final project during my undergraduate studies[1], I developed an automated trading bot based on an LSTM architecture.

It relied on a static approach: the AI made a prediction, and then a simple set of rules decided whether to buy or sell. While functional, this approach showed clear limitations in handling multiple assets and adapting to unpredictable market conditions.

This research aims to address those limitations by comparing this predictive method with Deep Reinforcement Learning, a more adaptive paradigm where an agent learns to trade by directly interacting with the market. The goal is to determine which system better handles multi-asset portfolios and volatile market conditions.

## 2 Description

The core aim of this research is to develop a comparative benchmarking framework that addresses the following limitations:

- **Data Pipeline:** the system will collect and process data from various assets, such as the Alpaca API or Yahoo Finance. This can be expanded by other data sources. However, the priority is to improve how the data quality is handled by creating a more efficient database to improve the training process for both the LSTM and the DRL agent.
- **Baseline Model (Predictive):** This part of the project focuses on improving and correcting elements of the previously developed LSTM model.

A major change will be converting the system from a "single-asset" to a "multi-asset" trading, allowing it to handle several stocks at once. This model will test several trading strategies to find which one works best for different types of assets.

Another key addition will be transforming the LSTM output from discrete trading signals (buy, sell, hold) into a probabilistic classification. Instead of directly generating actions, the model will estimate the likelihood of each possible outcome. This will introduce decision thresholds and improve the execution of trades, filtering out low-confidence predictions and reducing noise.

This probabilistic approach also aims to enable the integration of contextual filters (e.g., trend, momentum, or volatility conditions), and support more advanced decision policies, including risk-adjusted trade selection and dynamic position sizing. Overall, this change aims to improve both the robustness and interpretability of the system, while providing greater flexibility in strategy design and evaluation.

- **Challenger Model (Reactive):** The second model is an autonomous DRL agent, such as PPO or DDPG, that learns to trade by interacting directly with the market. The priority is to build a strong agent that works without sentiment data. Sentiment data will only be integrated if the core agent demonstrates stable performance without it.
- **Robustness and Stress-Testing module:** The single system to compare both models aims to use standard financial scores. The models will be evaluated under adverse market

conditions, including periods of high volatility and artificially injected noise. This will allow us to assess whether the model stays stable and protects capital better when the market becomes chaotic, in comparison with the previous model.

### 3 Objectives

| Priority  | Goals                                                                  |
|-----------|------------------------------------------------------------------------|
| Primary   | Design a robust data pipeline for multi-asset management.              |
|           | Improve and convert the previous LSTM model to handle multiple assets. |
|           | Set up the Reinforcement Learning environment.                         |
|           | Establish a benchmarking framework to compare both models.             |
| Secondary | Successfully train a DRL agent on multi-asset data.                    |
|           | Develop the Robustness and Stress-Testing module.                      |
| Tertiary  | Experiment with Sentiment Analysis as an optional data input.          |

### 4 Ethics

This project does not involve human participants, animal subjects, or sensitive personal data. It relies entirely on publicly available financial data and uses "Paper Trading" (simulated money) for testing.

### 5 Resources

| Category       | Requirements                                                                                                                                              |
|----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| Hardware       | Access to the School's GPU cluster and standard lab computers.                                                                                            |
| Software       | No specific license requests.                                                                                                                             |
| Data           | Already secured via personal API access[2]. Additional sources may be integrated, if necessary, to improve data quality. However, no funding is required. |
| Infrastructure | Private GitHub or GitLab repository for version control and code storage.                                                                                 |

## References

- [1] French RNCP37827 Certification - Available at: [\*https://www.francecompetences.fr/recherche/rncp/37827/\*](https://www.francecompetences.fr/recherche/rncp/37827/)
- [2] Alpaca Markets API - Available at: [\*https://alpaca.markets/\*](https://alpaca.markets/)